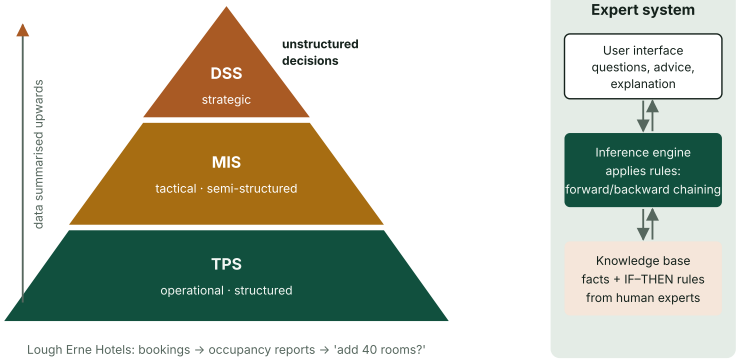
Types of information system

IS15 FACT FILE

Specification: students should be able to

- Describe the characteristics and purpose of transaction processing systems (TPS)
- Describe management information systems (MIS) and decision support systems (DSS)
- Describe expert systems, including the knowledge base, inference engine and user interface

Key words: information system, transaction processing system, batch processing, real-time processing, management information system, decision support system, what-if analysis, expert system, knowledge base, inference engine, explanation facility, strategic



The information pyramid: TPS at the operational level, MIS at the tactical level and DSS at the strategic level, with data flowing upwards; and the three components of an expert system

TPS and MIS

A transaction processing system captures and processes routine business transactions such as sales, bookings and payments, either in real time or in batches. It supports operational staff and provides the detailed data on which other systems depend. A management information system summarises TPS data into scheduled, exception and on-demand reports for middle managers, helping them monitor performance and make tactical, semi-structured decisions such as staffing levels or budgets.

Decision support systems

A decision support system helps managers tackle semi-structured and unstructured problems. It combines a database of internal and external data with a model base of analytical models and an interactive interface. Managers can carry out what-if analysis, goal seeking and forecasting to compare options. Unlike an MIS, which produces fixed reports, a DSS is flexible and interactive, but it supports rather than replaces human judgement.

Expert systems

An expert system captures the knowledge of human experts in a specific field. Its knowledge base holds facts and IF–THEN rules; its inference engine applies the rules to the facts supplied, using forward or backward chaining, to reach conclusions; its user interface asks questions and gives advice, often with an explanation facility. Expert systems give consistent, always-available advice but cannot use common sense, are costly to build and maintain, and raise questions of responsibility when wrong.

T Key terms from the mark schemes

Transaction processing system (TPS)
An information system that records and processes the routine day-to-day transactions of an organisation, in real time or in batches.

Management information system (MIS)
An information system that summarises data from transaction systems into regular, exception and on-demand reports for middle managers.

Decision support system (DSS)
An interactive information system that uses data and models to help managers make semi-structured and unstructured decisions, for example through what-if analysis.

Expert system
A computer system that emulates the decision-making of a human expert in a specific domain using a knowledge base and an inference engine.

Knowledge base
The part of an expert system that stores facts and rules gathered from human experts.

Inference engine
The part of an expert system that applies the rules in the knowledge base to the facts supplied in order to reach conclusions.

Exam focus

Questions often ask you to describe a system and apply it to a scenario, so state its purpose, its users and an example of its inputs and outputs in context. For expert systems, name all three components and say what each does – listing names alone rarely gains full marks. In discussion questions about expert systems, balance consistency and availability against the lack of common sense, cost and accountability, and conclude whether the system should support or replace the expert.

? Check yourself

1. State the difference between batch and real-time transaction processing, giving an example of each. [4]
2. Describe three types of report that a management information system might produce. [3]
3. Explain how a decision support system differs from a management information system. [3]

Answers: 1. Batch processing: transactions are collected together and processed as a group later, e.g. at night (1) / example: payroll or utility billing (1) / Real-time processing: each transaction is processed immediately and records are updated straight away (1) / example: cinema seat booking or bank withdrawal at an ATM (1). 2. Scheduled (periodic) reports produced at regular intervals, e.g. weekly sales by branch (1) / exception reports produced only when something falls outside normal limits, e.g. stores below target or hotel occupancy under 60% (1) / on-demand (ad hoc) reports produced when a manager requests them, e.g. a query on a specific product (1). Accept summary/detailed reports with example. 3. An MIS produces regular, structured reports, mostly from internal historical data, to support structured/semi-structured tactical decisions (1) / a DSS supports semi-structured or unstructured (often strategic) decisions using internal and external data and a model base (1) / it is interactive, allowing what-if analysis, goal seeking and forecasting, e.g. 'what happens to profit if fuel prices rise 10%?' (1).